

Kyla C. Reambonanza

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Education

Mindanao State University–Iligan Institute of Technology (MSU-IIT)

Iligan, PH

Bachelor of Science in Computer Science - Cum Laude (CGPA: 1.65/1.00)

July 2026

Relevant Coursework: Data Mining, Software Engineering, Human-Computer Interaction, Artificial Intelligence, Modelling and Simulation, Parallel and Computing

Publication & Research

Simulating the Spread of Antimicrobial Resistance in Agricultural Ecosystems Using an Agent-Based Approach with Parallelization

The 10th IEEE Cyber Science and Technology Congress (Peer-Reviewed) - Hokkaido, Japan (Online)

Oct. 2025

- Led the research and development of a parallelized agent-based simulation model using NetLogo and the pyNetLogo Python library to analyse how antimicrobial resistance (AMR) develops and transmits across agricultural environments to provide data-driven insights for global public health mitigation. [DOI: [10.1109/CyberSciTech68397.2025.00071](https://doi.org/10.1109/CyberSciTech68397.2025.00071)]

Bi-Objective Genetic Algorithm for Vessel Pickup Scheduling

Dec. 2024

- Formulated a bi-objective Genetic Algorithm using tournament selection, swap mutations, and elitism to optimise vessel routing and container capacity in a simulated port.

Technical Skills & Projects

Programming: Python, SQL, NetLogo, Laravel, HTML/CSS, Streamlit, Next.js

Design & Tools: Figma, Git, Bitbucket, Latex, Cloudinary, Jupyter Notebook

Language: Cebuano (Native), Tagalog (Native), English (CEFR C1), Chinese (Novice 2)

Real-Time Chicken Egg Crack Detection System: Undergraduate Thesis

[github/egg-shell-crack-detection](https://github.com/kayreambonanza/egg-shell-crack-detection)

- Researcher and developer of an automated system designed to detect cracks in chicken eggs utilizing Deep learning techniques, specifically Convolutional Neural Network (CNN) model using Python to process acoustic data transformed into Log-Mel Spectrograms.
- Designed an intuitive UI/UX featuring live inference results, confidence scores, spectrogram visual displays, and diagnostic logging for agricultural quality control using Streamlit.

Utility Pole Wire Hazard Assessment v2: Ongoing/Prototype

[github/electrical-post-clutter-segmentation-V2](https://github.com/kayreambonanza/electrical-post-clutter-segmentation-V2)

- Engineered an automated risk assessment pipeline using YOLOv8 instance segmentation to detect utility poles and dense wire clutters from urban street images.
- Trained custom segmentation weights on CUDA GPU hardware, achieving ~0.70 Mask mAP50 with early stopping optimization.
- Implemented an automated clutter severity metric and deployed a real-time OpenCV Heads-Up Display (HUD).

Professional Experience

Dotted Studios

Iligan, PH

UI/UX & Branding Intern

June 2025 – July 2025

- Designed intuitive, user-centered digital interfaces and visual brand systems based on accessibility guidelines and user research workflows.

Leadership & Activities

TechForKids: Teaching Digital Literacy in Far Fetched Areas

Iligan, PH

Volunteer Instructor

March 2026

- Collaborated with a team of volunteer educators to navigate logistical challenges and foster inclusive, encouraging learning environments for local underserved youth.

Computer Science Society

Iligan, PH

Undersecretary, Publication & Documentations Committee

June 2025 – May 2026

- Facilitated and supervised instructional support and assisted lead educators in delivering technical concepts to female officers, promoting digital and social inclusion.

College of Computer Studies Community Extension Program

Iligan, PH

Volunteer Teaching Assistant

March – June 2025

- Volunteered and taught the basics of Scratch to male PDLs (Persons Deprived of Liberty).